

The Optimization Landscape

When local information tells the global truth

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**The hook: two training runs,
two fates**

Same job, two mornings

Both of these minimize a loss — L14 taught us that's what “fitting” is:

	fit a linear model	train a deep network
the code	<code>np.linalg.lstsq(X, y)</code>	<code>loop of <code>loss.backward()</code> ; <code>opt.step()</code></code>
knobs to tune	none	learning rate, schedule, warmup, init, clipping, batch size, ...
rerun on the same data	same answer, every time	depends on the seed
failure modes	—	divergence, plateaus, nan at step 617 (L2!)

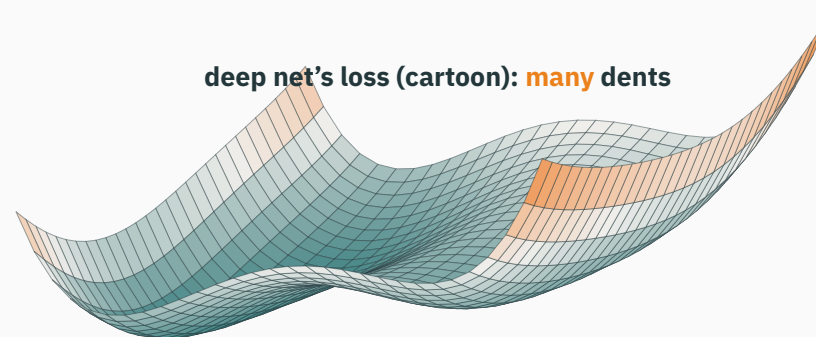
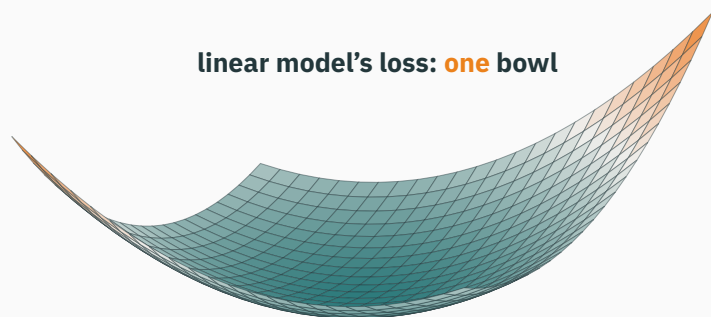
Same species of problem — **minimize a function of the parameters** — wildly different lives.

Same job, two mornings

Commit to a guess: is the difference in the *algorithms*, the *hardware*, or the *function being minimized*? Today's lecture is the answer.

The answer, before any definitions

The difference is the **shape of the surface** each one walks on:



- Left: wherever you start, downhill leads to **the** bottom — no tricks required
- Right: valleys, ridges, saddles — where you end up **depends on where you start**

Today's word for the left picture is **convex. Convexity is the property that makes “just run the optimizer” a mathematically safe instruction.**

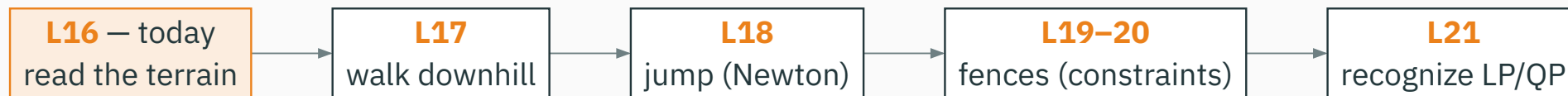
Module 3 shook hands with Module 4

L14 closed with a promise — every fit is a minimization:

$$\hat{\theta} = \arg \min_{\theta} \underbrace{\text{NLL}(\theta)}_{\text{MSE, BCE, CE, ...}}$$

L14: “Fitting = minimizing NLL. Module 4 is the art of walking downhill

Module 4 keeps that promise, one lecture per skill:



Before anyone walks anywhere: learn to **read the terrain**. That is today.

Learning outcomes

By the end of this lecture you will be able to:

1. Use the optimizer's vocabulary: **objective, minimizer, stationary point, local vs global**.
2. Classify flat spots by the Hessian's **eigenvalue signs** — L9's gallery, now on the job.
3. Test a **set** for convexity with segments, and a **function** with chords.
4. Certify convexity fast: $f'' \geq 0 / H \succeq 0$, plus the **sum, max, affine-composition** rules.
5. Prove the module's license plate (★): **convex $\Rightarrow$ every local minimum is global**.
6. Prove **least squares is convex** via $H = 2X^\top X \succeq 0$ (★) — and say what deep nets give up.

The optimizer's predicament

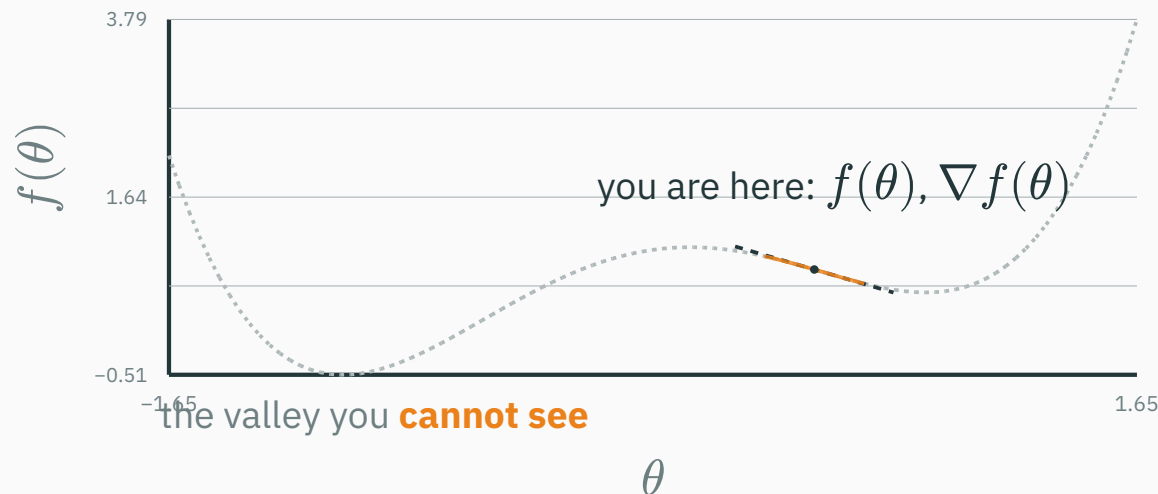
The problem, stated once for the whole module

$$\min_{\theta \in \mathbb{R}^d} f(\theta)$$

- f : the **objective** — for us, almost always a loss / NLL (L14)
- a **minimizer** θ^* : an input where the smallest value is reached — $\theta^* = \arg \min_{\theta} f(\theta)$; the **minimum** $f(\theta^*)$ is the value there
- maximizing is the same craft upside-down: $\max f = -\min(-f)$ — L14 already negated log-likelihoods for exactly this reason, so **everything from now on is a minimization**

One problem shape rules all of ML: $\min_{\theta} f(\theta)$. The whole module is about this line.

You never see the landscape — only a flashlight beam



Standing at θ , the fog gives you three legal questions — nothing else:

probe	what it is in ML	cost
$f(\theta)$	run the model, read the loss	cheap
$\nabla f(\theta)$	backprop (L8–L11)	cheap — one backward pass
$H(\theta)$	curvature matrix (L9)	d^2 entries — rarely affordable

Why not just look everywhere?

Because θ lives in $\mathbb{R}^d$, and d is not 2. Grid-search with a modest 10 values per coordinate:

model	d	grid evaluations 10^d
line $wx + b$	2	$10^2 = 100$ — trivial
tiny MLP	10^3	10^{1000} — no.
GPT-class model	10^{11}	$10^{10^{11}}$ — the observable universe has $\sim 10^{80}$ atoms

At ML scale, local probes are all you will ever have. Optimization is the art of acting globally on local information.

This is the module's recurring tension: the flashlight is honest but **near-sighted**. Today we find the one world where near-sighted is enough.

What the probes buy: Taylor, the flashlight's optics

L9 built “the formula every optimizer lives on” — today it starts living:

$$f(\theta + \delta) \approx \underbrace{f(\theta)}_{\text{probe 1}} + \underbrace{\nabla f(\theta)^\top \delta}_{\text{probe 2: the local plane}} + \underbrace{\frac{1}{2} \delta^\top H(\theta) \delta}_{\text{probe 3: the local bowl or saddle}}$$

- L17 trusts the first two terms for one small step — **gradient descent**
- L18 trusts all three and jumps — **Newton's method**
- Today's question is prior to both: when do **local** conclusions (“it's flat here”, “it curves up here”) say anything about the **global** landscape?

**Stationary points: where the
search can stop**

Where can a local method stop?

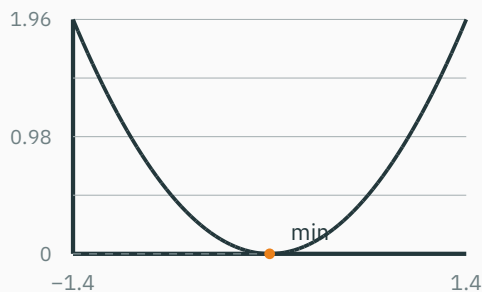
While $\nabla f(\theta) \neq \mathbf{0}$, the Taylor plane is tilted — a small step against the gradient strictly lowers f (L17 makes this precise). So a sensible walker **keeps moving**.

it can only park where $\nabla f(\theta) = \mathbf{0}$ — a **stationary point**

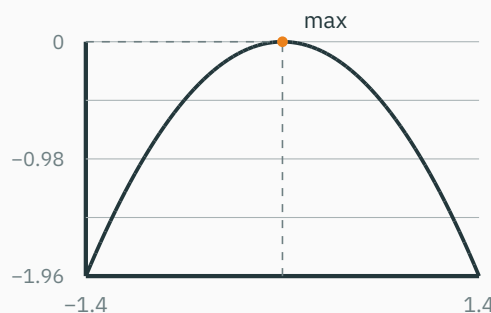
- Also called a **critical point**; the flashlight sees “flat” there
- But **flat is not the same as bottom** — L7 met smiles, frowns and x^3 ’s fake-out in 1-D

Optimizers are moths: they can only land on flat spots. The rest of this section is a field guide to flat spots.

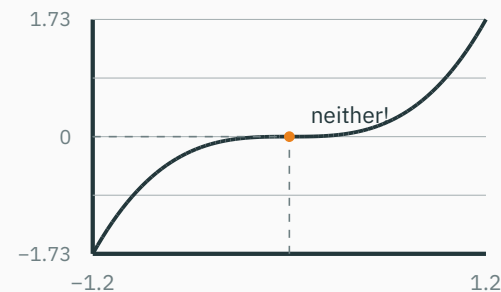
The 1-D field guide, in thirty seconds



$f'' = 2 > 0$ – smile



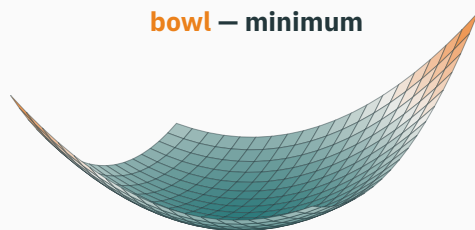
$f'' = -2 < 0$ – frown



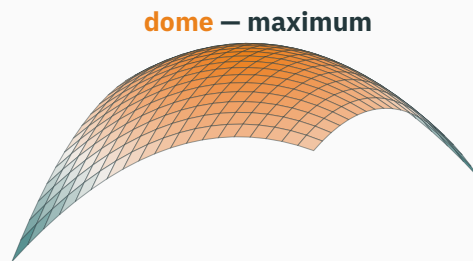
$f'' = 6x = 0$ – no verdict

All three have $f'(0) = 0$. The **second** derivative tells them apart — except when it's 0, where the test goes silent (x^3 : neither a min nor a max).

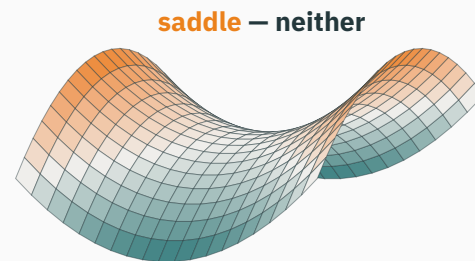
In d dimensions the field guide is L9's gallery



$$\lambda = 3, 1 - \text{all} > 0$$



$$\lambda = -1, -3 - \text{all} < 0$$



$$\lambda = 2, -2 - \text{mixed signs}$$

Curvature became a matrix in L9; its **eigenvalue signs** are the shape. (Eigenvalues above: computed live by the in-slide solver.)

At a stationary point: H 's eigenvalue signs name the flat spot — all + min, all — max, mixed = saddle.

The verdict table, ready for work

eigenvalues of H at $\nabla f = 0$	verdict	what a minimizer should do
all $\lambda_i > 0$ (PD)	local minimum	park — locally, done
all $\lambda_i < 0$ (ND)	local maximum	any direction escapes
mixed signs (indefinite)	saddle	escape along a $\lambda < 0$ eigenvector
some $\lambda_i = 0$	test is silent	look beyond second order (x^3 , troughs)

Hand tests for 2×2 still apply (L9): $\det H < 0 \rightarrow$ saddle; $\det H > 0$ — read the trace.

The verdict is **local**. “All $\lambda > 0$ ” certifies a neighborhood, not the world — hold that thought; it is today’s entire plot.

Worked numbers · two flat spots, two fates

$$f(x, y) = x^3 - 3x + y^2 \quad \nabla f = \begin{pmatrix} 3x^2 - 3 \\ 2y \end{pmatrix} = \mathbf{0} \Rightarrow (x, y) = (\pm 1, 0)$$

Hessian: $H = \begin{pmatrix} 6x & 0 \\ 0 & 2 \end{pmatrix}$ — depends on where you stand (curvature is local):

- At $(1, 0)$: $H = \begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$, $\lambda = 6, 2$ — both positive → **local minimum**
- At $(-1, 0)$: $H = \begin{pmatrix} -6 & 0 \\ 0 & 2 \end{pmatrix}$, $\lambda = -6, 2$ — mixed → **saddle**

And since $f(x, 0) = x^3 - 3x \rightarrow -\infty$ as $x \rightarrow -\infty$: that “local minimum” is **not** global — nothing is. Flat-spot analysis alone can never tell you this.

Code: the field guide in six lines

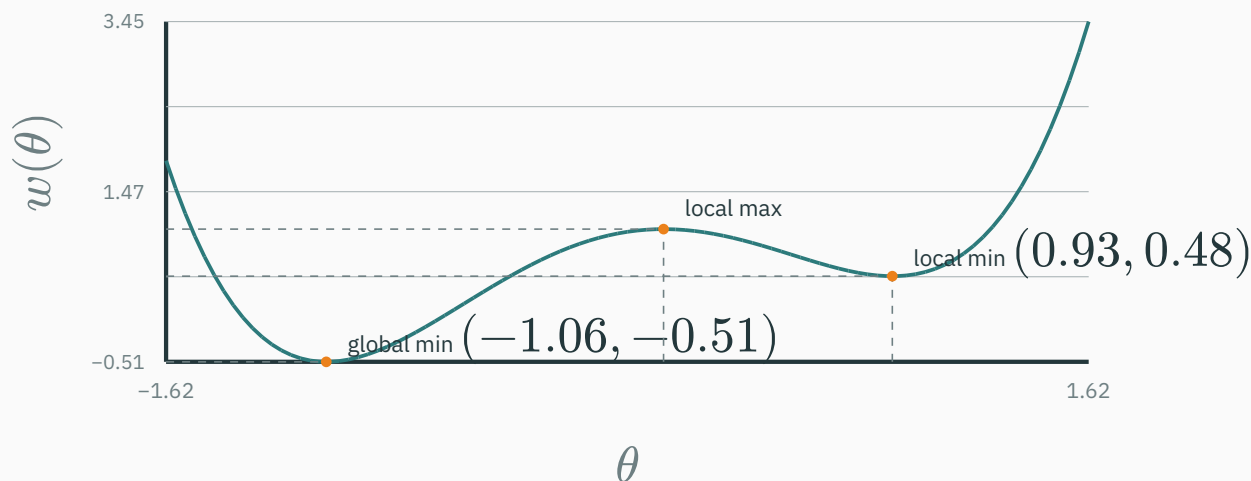
```
import numpy as np

def hess(x, y):
    # for f = x**3 - 3*x + y**2
    return np.array([[6*x, 0.0], [0.0, 2.0]])

for p in [(1, 0), (-1, 0)]:
    lam = np.linalg.eigvalsh(hess(*p))
    # symmetric -> real, sorted
    kind = "min" if lam[0] > 0 else ("max" if lam[-1] < 0 else "saddle")
    print(p, lam, kind)
# (1, 0) [2. 6.] min
# (-1, 0) [-6. 2.] saddle
```

eigvalsh is the same three-line ritual as L9 — the new part is what the optimizer **does** with the verdict.

Our non-convex mascot for the day, $w(x) = (x^2 - 1)^2 + x/2$ — every marked value below is computed in-slide (Newton on $w' = 0$):



- **Local minimum**: no better point in some ball around it. **Global minimum**: no better point anywhere.
- A walker started at $\theta = 1.5$ parks at 0.93 — flat, curving up, **fully convinced** — and is wrong by 1.00 in loss.

No local probe at 0.93 — not f , not ∇f , not H — betrays the better valley at -1.06 .
The flashlight cannot call its bluff.

Checkpoint: what does the flashlight certify?

QUESTION

An optimizer stops at $\hat{\theta}$ on a smooth, **unknown** $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, with $\nabla f(\hat{\theta}) = 0$ and $H(\hat{\theta})$ having eigenvalues 2 and 5. What is **certain**?

A. $\hat{\theta}$ is a local minimum — and that is all we can certify

B. $\hat{\theta}$ is the global minimum

C. $\hat{\theta}$ is a saddle point

D. f is convex

Correct: A — a local minimum — and nothing more

Why: Both eigenvalues positive $\rightarrow$ PD $\rightarrow$ a genuine local min (L9). But **global** (B) is a claim about all of $\mathbb{R}^2$, and local probes at one point can't reach that far — our two-valley mascot just demonstrated it. (D) reverses today's logic: curvature at **one** point never certifies convexity — x^3 has $f''(1) = 6 > 0$ yet isn't convex. Turning “local min” into “global min” needs a **property of f** , not a better probe. Enter convexity.

**Convex sets: segments that
never leave**

The fix must be a property of shapes

What we want is a **certificate**:

$$\underbrace{\text{local verdict}}_{\nabla f=0, H \succeq 0} + \underbrace{\text{a property of } f \text{ as a whole}}_{?} \Rightarrow \text{global claim}$$

That property is **convexity**, and it comes in two pieces, buildable in one lecture:

- **convex sets** — regions where straight segments never leave (this section)
- **convex functions** — surfaces whose chords never dip below the graph (next)

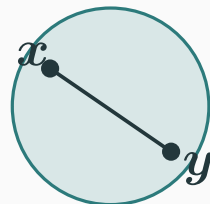
Sets first because functions need somewhere to live: the chord test only makes sense if the segment between two inputs stays inside the domain.

The segment test

A set $C \subseteq \mathbb{R}^d$ is **convex** if for all $x, y \in C$ and every $t \in [0, 1]$:

$$tx + (1 - t)y \in C$$

- $tx + (1 - t)y$ is a point **on the segment** from y to x — t is the slider
- Convex = “walk between any two members and you never step outside”



One definition, one picture: no shortcut between members ever leaves the club.

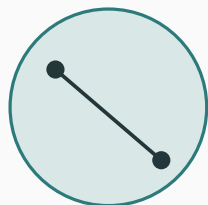
Numbers: walking the segment

Take $x = (4, 1)$ and $y = (0, 3)$. The mix $tx + (1 - t)y$, by hand:

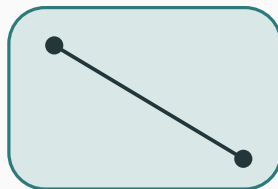
t	0	1/4	1/2	3/4	1
mix	(0, 3)	(1, 2.5)	(2, 2)	(3, 1.5)	(4, 1)

- $t = 0$ sits at y , $t = 1$ at x ; in between, a straight walk — each coordinate is mixed independently
- The same expression with weights $(t, 1 - t)$ will reappear within minutes as the **chord** of a function — same algebra, one dimension up

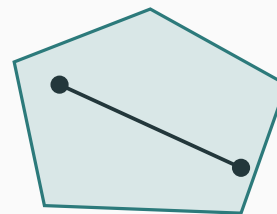
Vocabulary you'll meet in Boyd: a **convex combination** is a mix with nonnegative weights summing to 1 — segments are the 2-point case.



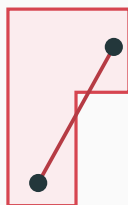
disk ✓



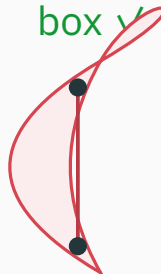
box ✓



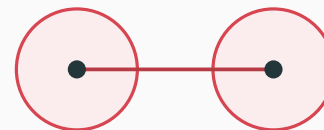
polytope ✓



a notch ✗



a crescent ✗



two islands ✗

One escaped segment is enough to convict: convexity is a **for-all** claim, non-convexity needs one witness.

The convex sets ML lives on

set	where you meet it	convex?
$\mathbb{R}^d$ — all of parameter space	unconstrained training	✓
box $[0, 1]^d$	valid pixel intensities	✓
half-space $\{x : a^\top x \leq b\}$	one side of a flat wall	✓
norm ball $\{\theta : \ \theta\ \leq r\}$ (L3)	weight constraints, regularization	✓
simplex $\{p : p_i \geq 0, \sum_i p_i = 1\}$	probability vectors — softmax outputs live here	✓ (L19 optimizes over it)
sphere $\{x : \ x\ = 1\}$ — the skin only	normalized embeddings	✗

The sphere fails beautifully: mix two antipodes at $t = 1/2$ and land at 0 — the center, off the skin. **Ball convex, shell not.**

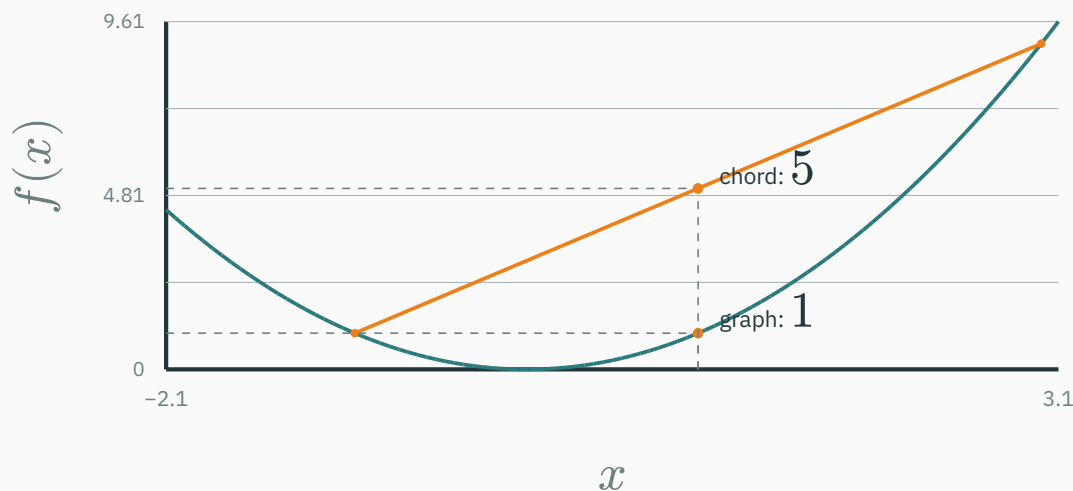
The convex sets ML lives on

Intersections preserve convexity — clip convex sets together and the result stays convex (each fence is respected). L21's LP feasible regions are exactly such stacks of half-spaces. Unions, as the two islands showed, do not.

Convex functions: the chord test

Lift the segment to the graph

A function is **convex** when every chord — the segment joining two points **on the graph** — sits on or above the graph:



$$f\left(\underbrace{tx_1 + (1-t)x_2}_{\text{mix the inputs}}\right) \leq \underbrace{tf(x_1) + (1-t)f(x_2)}_{\text{mix the outputs = chord height}} \quad \forall x_1, x_2, t \in [0, 1]$$

Convex: the graph hangs below its chords. (Domain must be a convex set
— now you know why sets came first.)

Reading the inequality with honest numbers

$f(x) = x^2$, endpoints $x_1 = -1$, $x_2 = 3$. Mixed input: $tx_1 + (1 - t)x_2 = 3 - 4t$.

t	0	1/4	1/2	3/4	1
mixed input $3 - 4t$	3	2	1	0	-1
graph $f(\text{mix}) = (3 - 4t)^2$	9	4	1	0	1
chord $tf(-1) + (1 - t)f(3) = 9 - 8t$	9	7	5	3	1

Graph $\leq$ chord in every column — with equality exactly at the endpoints. That is the definition, felt in integers.

Mnemonic: conve*x **like a cu*p** — no. Try this instead: convex holds water (U-shaped), concave is a cave (∩-shaped). Crude, universal, effective.

Two checks that need no calculus

$f(x) = x^2$, **by pure algebra** (the chord inequality, expanded):

$$\text{LHS} - \text{RHS} = (tx_1 + (1-t)x_2)^2 - tx_1^2 - (1-t)x_2^2 = -t(1-t)(x_1 - x_2)^2 \leq 0 \quad \checkmark$$

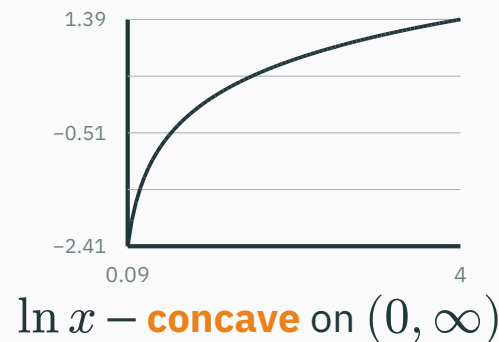
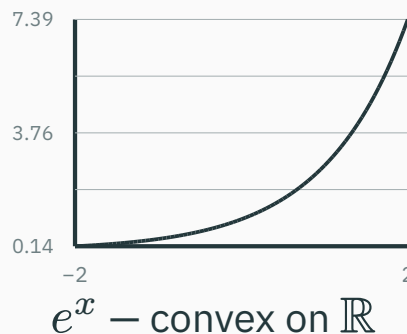
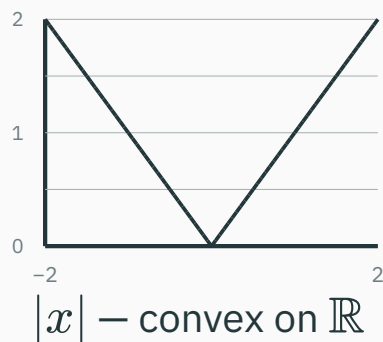
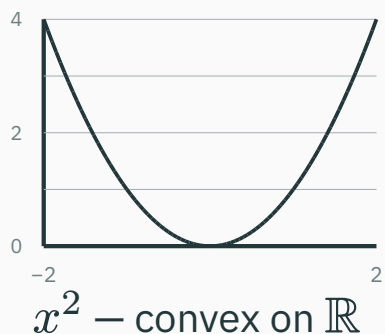
($t(1-t) \geq 0$ on $[0, 1]$; a square is ≥ 0 . Multiply out once at home — three lines.)

$f(x) = |x|$, **by the triangle inequality** (L3):

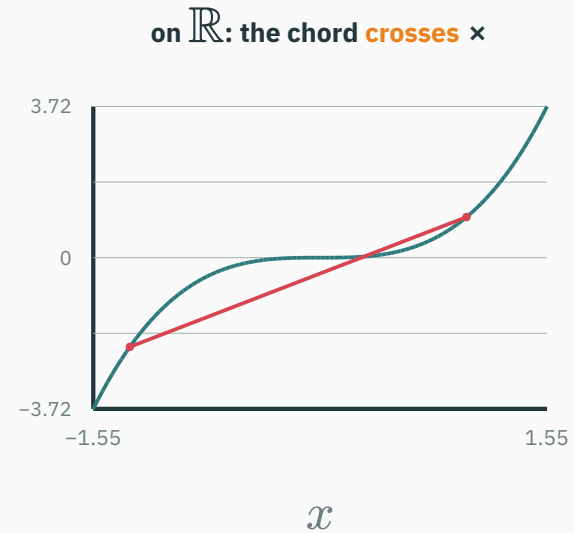
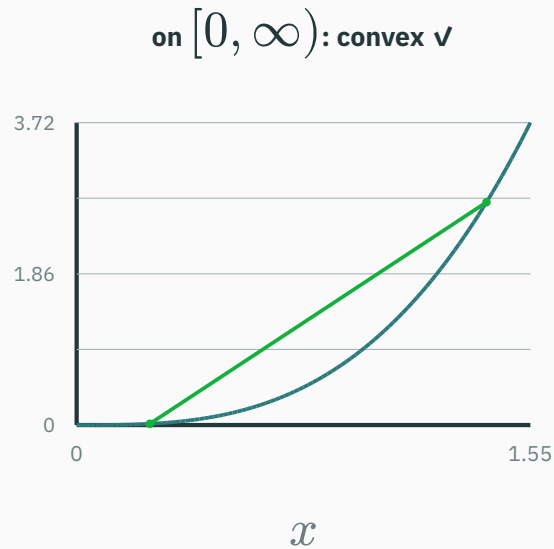
$$|tx_1 + (1-t)x_2| \leq |tx_1| + |(1-t)x_2| = t|x_1| + (1-t)|x_2| \quad \checkmark$$

The chord definition never mentions derivatives — it happily certifies the **kinked $|x|$, where $f''(0)$ does not even exist.**

The 1-D gallery, with verdicts



- e^x : convex even though it never turns around — convexity is about **bending up**, not about having a minimum
- $\ln x$: bends **down** everywhere — concave; and $-\ln x$ is convex (the shape of every NLL's building block, L14)



- On $[0, \infty)$ every chord clears the graph; on $(-\infty, 0]$ it's concave; on all of $\mathbb{R}$, neither — the right chord dips below **and** rises above
- “Is f convex?” is ill-posed until you say **on which (convex) set** — same lesson the optimizers will learn about where they are allowed to search

Concave: the same story upside-down

f is **concave** $\Leftrightarrow -f$ is convex $\Leftrightarrow$ chords hang **below** the graph.

Why we care — Module 3 was secretly a concavity story:

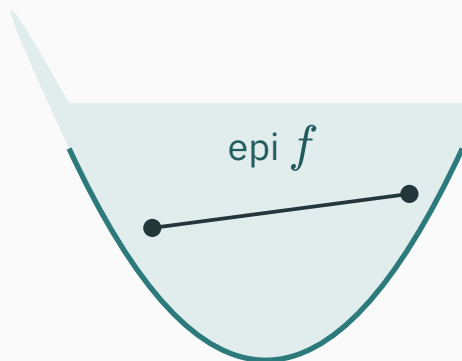
$$\underbrace{\max_{\theta} \ell(\theta)}_{\text{maximize log-likelihood (L14)}} = \underbrace{\min_{\theta} \text{NLL}(\theta)}_{\text{minimize a convex loss — often}}$$

- The coin's $\ell(\theta) = 7 \ln \theta + 3 \ln(1 - \theta)$: sum of concaves — concave; its NLL is convex
- Gaussian $\ell(\mu)$: a downward parabola in μ (L14 showed it) — concave; MSE convex

Maximizing a concave likelihood **is minimizing a convex loss. Module 3's climbs were secretly walks in a bowl.**

One idea, two costumes: the epigraph

The two halves of today are the same object. The **epigraph** of f is everything on or above the graph:



$$\text{epi } f = \{(x, v) : v \geq f(x)\}$$

- f is a convex **function** $\Leftrightarrow$ epi f is a convex **set**
- A chord's endpoints are in the epigraph, so the whole chord must be — and “chord stays in epi f ” **is** “chord above graph”

This is Boyd's favourite bridge (Ch 3.1.7) — one segment test runs the entire theory. Nothing new to memorize.

Recognizing convexity without chords

Chords are the definition — calculus is the shortcut

Checking chords for every pair is honest but slow. For twice-differentiable f , curvature settles it:

f convex on an interval $\Leftrightarrow f''(x) \geq 0$ everywhere on it

The gallery, re-certified in one line each:

f	f''	verdict
x^2	$2 > 0$	convex on $\mathbb{R}$ (and strictly — no flat stretches)
e^x	$e^x > 0$	convex on $\mathbb{R}$
$\ln x$	$-1/x^2 < 0$	concave on $(0, \infty)$
x^3	$6x$ — changes sign	convex on $[0, \infty)$ only
$ x $	undefined at 0	test silent — the chord already convicted $\vee$ convex

In d dimensions: the Hessian test

f convex $\Leftrightarrow H(\theta) \succeq 0$ (PSD) for **every** θ L9's "official words", now a tool

Worked: $f(\theta_1, \theta_2) = \theta_1^2 + \theta_2^2$.

$$H = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \text{ everywhere} \Rightarrow \lambda = 2, 2 > 0 \Rightarrow \text{convex (strictly)} \checkmark$$

And the whole quadratic family at once: $q(\theta) = \frac{1}{2}\theta^\top A\theta$ has $H = A$ everywhere (L9's ★) — so **q convex $\Leftrightarrow A \succeq 0$** : bowls and troughs yes, saddles and domes no. The shape gallery was a convexity gallery all along.

The test demands PSD at **every** point of a convex domain — one PD Hessian at one point proves nothing (x^3 at $x = 1$, again).

The composition rules: convexity is stackable

Convex pieces snap together — for f, g convex:

rule	why (one breath)
αf for $\alpha \geq 0$	scaling both sides of the chord inequality preserves it
$f + g$	add the two chord inequalities
$f(Ax + b)$ — affine into convex	an affine map sends segments to segments, then f 's chords take over
$\max(f, g)$	the upper envelope: each chord clears whichever graph is on top

What the table does **not** say: products and general compositions break it. Star witness: $(x - 1)^2 \cdot (x + 1)^2 = (x^2 - 1)^2$ — two convex parabolas multiply into **our**

The composition rules: convexity is stackable

double-well mascot. Non-convexity is what multiplication (and layer-stacking!) does.

Assemble the objectives you'll actually meet

objective	build	convex?
$\ X\theta - \mathbf{y}\ ^2$ — least squares	affine $\theta \mapsto X\theta - \mathbf{y}$ into $\ \cdot\ ^2$	✓ (★ proof next section)
$\ X\theta - \mathbf{y}\ ^2 + \lambda\ \theta\ ^2$ — ridge	sum of two convex terms	✓
$\ X\theta - \mathbf{y}\ ^2 + \lambda \sum_i \theta_i $ — lasso	sum; $ \cdot $ of an affine map	✓
$\max(0, 1 - z(\theta))$, z affine — hinge	max of affine and constant	✓ (SVMs, ML course)
deep net loss	compositions upon compositions	✗ in general

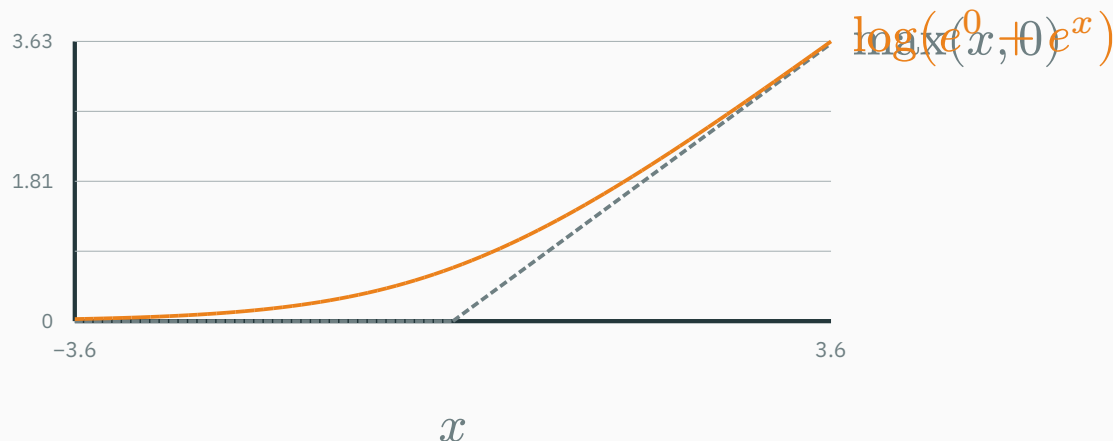
Regularizing **cannot break** convexity — L15's “every regularizer is a prior” now has an optimization corollary: priors are free, geometrically.

Assemble the objectives you'll actually meet

You will almost never chord-check a real objective. You will **assemble it from certified parts — this table is how convexity scales.**

Log-sum-exp is convex — L2's hero returns

L2 derived $\text{LSE}(x) = \log \sum_i e^{x_i}$ for numerical survival. It is also **convex** — here in 1-D against the max it approximates:



- A **smooth max**: hugs $\max(x, 0)$, never kinks — max's convexity survives the smoothing
- Why convex: its Hessian is $H = \text{diag}(p) - pp^\top$ with $p = \text{softmax}(x)$, and $v^\top H v = \text{Var}_{i \sim p}[v_i] \geq 0$ — **a variance** (L12) cannot be negative

Log-sum-exp is convex — L2's hero returns

Consequence you'll cash in ML: logistic-regression training minimizes LSE — affine, so it is convex — that's why it “just works” like least squares does.

Which one is **not** convex on all of $\mathbb{R}^2$, $\theta = (\theta_1, \theta_2)$?

A. $\|X\theta - y\|^2 + 5\|\theta\|^2$

B. $\max(\theta_1 + \theta_2, e^{\theta_1})$

C. $\log(e^{\theta_1} + e^{\theta_2})$

D. $\theta_1^2 - \theta_2^2$

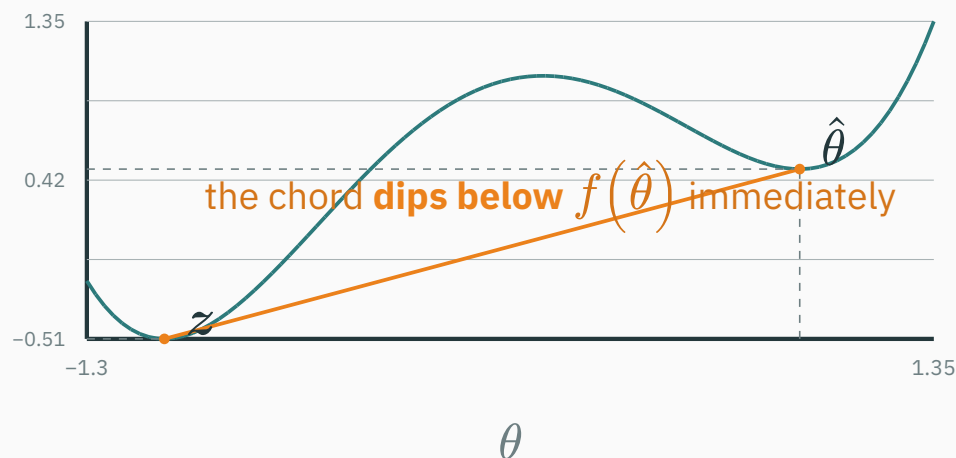
Correct: D — $\theta_1^2 - \theta_2^2$ — a saddle, not a bowl

Why: Its Hessian is $\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$ everywhere — mixed eigenvalue signs, indefinite, the gallery's saddle: not convex (and not concave). The rest assemble from certified parts: (A) sum of convex $\checkmark$; (B) max of an affine and a convex $\checkmark$; (C) log-sum-exp $\checkmark$. Note how you convicted D and certified A–C **without one chord** — the toolkit is the point.

**The payoff: local = global — and
least squares**

If f is convex, **every local minimum is a global minimum.**

Proof by picture first — suppose a convex f had a local min $\hat{\theta}$ **and** a strictly better point z :



The chord from $(\hat{\theta}, f(\hat{\theta}))$ down to $(z, f(z))$ starts sloping **down**. Convexity forces the graph to stay **under** that chord — so arbitrarily close to $\hat{\theta}$ there are strictly lower

★ The license plate of convex optimization

D DERIVATION

points. “Local minimum” was a lie. (Our mascot is non-convex — exactly why it **can** hide z behind a hill.)

★ The same picture, in four lines of algebra

Setup. f convex; $\hat{\theta}$ a local min; suppose some z has $f(z) < f(\hat{\theta})$.

Walk toward z . For $t \in (0, 1]$, the mix $\theta_t = (1 - t)\hat{\theta} + tz$ lies on the segment.

Convexity:

$$f(\theta_t) \leq (1 - t)f(\hat{\theta}) + tf(z)$$

Use the bad news. Since $f(z) < f(\hat{\theta})$:

$$f(\theta_t) < (1 - t)f(\hat{\theta}) + tf(\hat{\theta}) = f(\hat{\theta})$$

Contradiction. As $t \rightarrow 0$, θ_t enters **every** ball around $\hat{\theta}$ — yet each θ_t is strictly lower. No ball certifies $\hat{\theta}$ as a local min. ■

★ The same picture, in four lines of algebra

D DERIVATION

Convexity upgrades the flashlight: any parking spot it certifies **locally is certified **globally**. This theorem is the license the next five lectures drive on.**

★ Least squares, the star customer

D DERIVATION

The objective L4 set up, L6 solved at scale, L14 crowned as Gaussian NLL:

$$f(\theta) = \|X\theta - \mathbf{y}\|^2 = (X\theta - \mathbf{y})^\top (X\theta - \mathbf{y}) = \theta^\top X^\top X\theta - 2\mathbf{y}^\top X\theta + \mathbf{y}^\top \mathbf{y}$$

Differentiate with L9's two earned rules ($\nabla \theta^\top A\theta = 2A\theta$ for symmetric A ; $\nabla \mathbf{b}^\top \theta = \mathbf{b}$):

$$\nabla f(\theta) = 2X^\top X\theta - 2X^\top \mathbf{y} \quad H = \nabla(\nabla f) = 2X^\top X \quad \text{--- constant: the same bowl everywhere}$$

Convexity now hangs on one question: is $X^\top X$ **positive semi-definite for every possible data matrix** X ?

★ The one-line certificate: $X^\top X \succeq 0$, always

D DERIVATION

Take any direction $v \in \mathbb{R}^d$ and test the quadratic form (L9's definition of PSD):

$$v^\top (X^\top X) v = (Xv)^\top (Xv) = \|Xv\|^2 \geq 0 \quad \checkmark$$

A squared length cannot be negative — that is the entire proof. $H = 2X^\top X \succeq 0$ for any data, so least squares is convex: its stationary point is **the global minimum.**

On our tiny dataset $x = (0, 1, 2)$, $y = (1, 2, 4)$ — everything below computed live in-slide:

$$X^\top X = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \quad \lambda = 7.16, 0.84 > 0, \quad \nabla f = 0 \Rightarrow \theta^* = \begin{pmatrix} 0.83 \\ 1.50 \end{pmatrix}$$

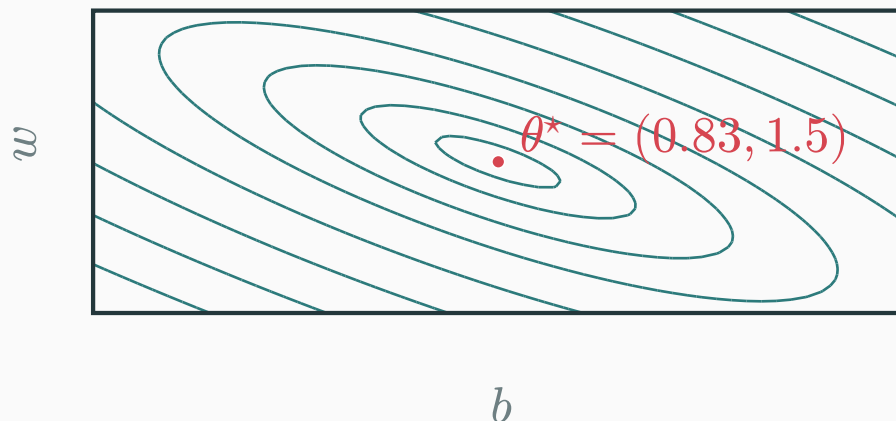
★ The one-line certificate: $X^\top X \succeq 0$, always

D DERIVATION

Both eigenvalues strictly positive $\rightarrow$ strictly convex $\rightarrow$ that θ^* ($\hat{y} = 1.5x + 0.83$) is **the unique best line in the universe** for this data. Certified.

The certified bowl, seen from above

$f(b, w) = \sum_i (b + wx_i - y_i)^2$ over the whole parameter plane — one basin, one bottom:



- Elongated, tilted — but **one** valley: every downhill path drains to θ^* (L17 walks it; the tilt and stretch will matter there)
- When columns of X are linearly dependent (L4's rank!), an eigenvalue of $X^\top X$ hits 0: the bowl grows a **flat trough** of equally-good minimizers — still convex, still all-global, just no longer unique

The hook, closed

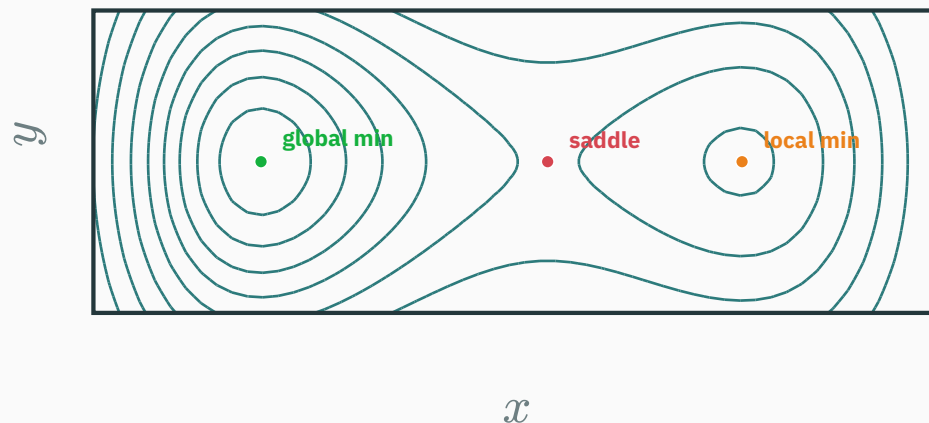
Why does the linear model “just work”?

- Least squares (and ridge, lasso, logistic regression — today’s toolkit) are **convex**: one basin, local = global, any reasonable walker succeeds — no seeds, no warmup, no babysitting
- A deep network is **compositions of compositions** — the one operation the rule table refuses to bless. Its landscape genuinely holds valleys, ridges, saddles, plateaus
- So deep learning’s “tricks” are not superstition — they are **navigation gear for non-convex terrain**

Convexity is not a luxury; it is the boundary between “guaranteed” and “we’ll see”. Know which side your problem lives on.

**Life without the promise — and
the road ahead**

Our mascot's 2-D sibling, $R(x, y) = (x^2 - 1)^2 + x/2 + 1.2y^2$ — all three flavors of flat spot in one place (marks placed by the in-slide Newton solver):



- The 1-D local **max** at 0.13 became a 2-D **saddle** (down along x , up along y) — extra dimensions turn maxima into saddles
- Two basins, one ridge between: where you land is decided by where you start

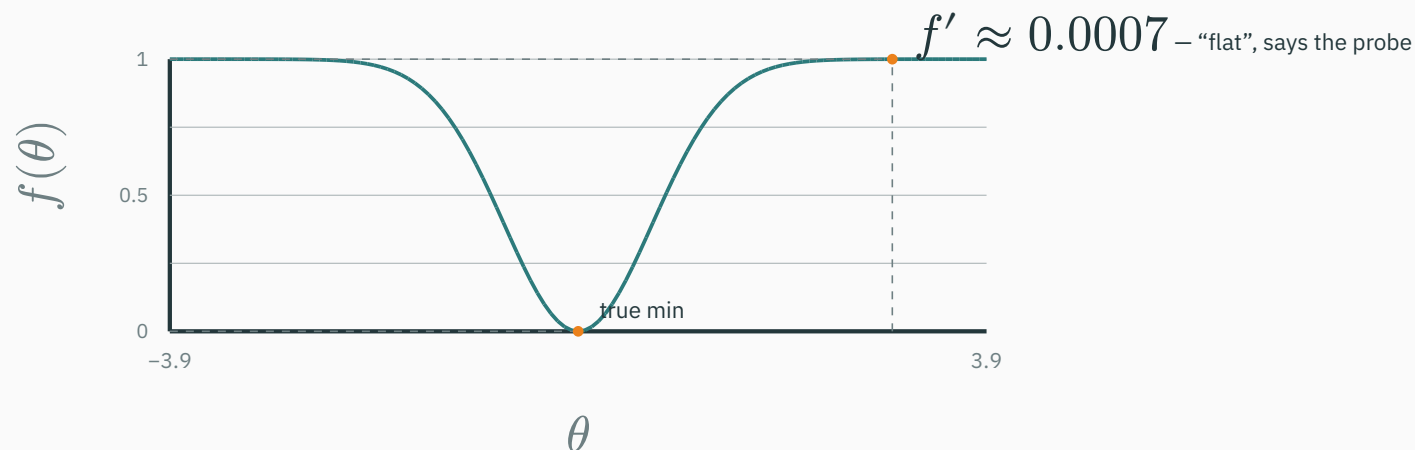
In high dimensions, saddles are the rule

Heuristic: at a random stationary point, each eigenvalue's sign is roughly a coin flip.
All d positive at once?

d	$P(\text{all } \lambda > 0) \approx 2^{-d}$	meaning
2	0.25	minima common in toy problems
20	$\approx 10^{-6}$	minima already rare
10^6 (a small net)	$\approx 10^{-301030}$	a random flat spot is a saddle, essentially always

- Good news hiding inside: a saddle has a $\lambda < 0$ eigen-direction — a visible downhill escape (L17–L18 exploit it); saddles **slow** walkers, they rarely trap them
- The heuristic is honest hand-waving; random-matrix theory makes it rigorous (Dauphin et al., 2014 — DL course territory)

Plateaus: where the flashlight goes dark



- On a **plateau**, $\nabla f \approx 0$ with no minimum anywhere near: steps shrink to a crawl — the failure mode behind “training is stuck” (and behind saturated activations, as the DL course will show)
- Yet deep learning works. Empirically: with enough parameters, most local minima are **nearly as good as the global one**, and modern initialization/normalization smooth the terrain. Facts used daily, proved nowhere near this course — honest IOU to ES 667

Plateaus: where the flashlight goes dark

Perspective: convexity is the **theory** of trustworthy landscapes; deep learning is the **engineering** of livable non-convex ones. You now own the vocabulary both sides negotiate in.

Lecture 16 — summary

- **One problem shape:** $\min_{\theta} f(\theta)$ — and you only ever hold local probes $f, \nabla f, H$ at your current θ (Taylor is the flashlight).
- **Stationary points** $\nabla f = 0$: classified by H 's eigenvalue signs (L9) — bowl / dome / saddle; the verdict is local only.
- **Convex set:** segments never leave. **Convex function:** chords never dip below the graph; epigraph ties the two.
- **Certify fast:** $f'' \geq 0$ / $H \succeq 0$ everywhere; or assemble from rules — scale, sum, affine composition, max; log-sum-exp is convex (a variance says so).
- ★ **Convex $\Rightarrow$ local = global** — the four-line chord contradiction.
- ★ **Least squares is convex:** $H = 2X^{\top}X$ and $v^{\top}X^{\top}Xv = \|Xv\|^2 \geq 0$ — for any data.
- **Without convexity:** saddles rule high dimensions, plateaus stall walkers — deep learning's daily weather, managed by tricks, not theorems.

Lecture 16 — summary

Read before L17 — MML §7.3 to 7.3.1 (convexity, 4 pages, our notation) · Boyd & Vandenberghe, *Convex Optimization* (free PDF): **skim** §2.1–2.3 for set pictures and §3.1.1–3.1.5 + 3.2 for the function tests and operation rules — read figures and boxed statements, skip proofs on a first pass. The Distill momentum article waits for L17.

Practice problems

Try on paper; anything numeric can be checked with five lines of NumPy.

1. Chord-check $f(x) = |2x - 1|$: prove convexity from the definition (triangle inequality + the affine trick).
2. Find **all** stationary points of $f(x, y) = x^4 + y^4 - 4xy$ and classify each with the Hessian. Which are global? (Careful: there's a tie.)
3. Is $g(x) = xe^x$ convex on $\mathbb{R}$? On which interval is it convex? ($g'' = (x + 2)e^x$.)
4. Ridge: show $H = 2X^\top X + 2\lambda I$ and explain why $\lambda > 0$ makes H positive **definite** even when $X^\top X$ has a zero eigenvalue. What does that do to the trough of equally-good minimizers?
5. Prove or refute with a picture: (a) the intersection of two convex sets is convex; (b) the union is.
6. For $d = 2$, evaluate log-sum-exp's Hessian $\text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top$ at $\mathbf{x} = (0, 0)$, find the eigenvector with $\lambda = 0$, and connect it to L2's subtract-the-max trick.

★★★ Jensen's inequality: chords, industrialized

★ OPTIONAL

Run the chord inequality with n points instead of 2 (induction), weights $t_i \geq 0$, $\sum t_i = 1$:

$$f\left(\sum_i t_i x_i\right) \leq \sum_i t_i f(x_i) \quad \text{and with } t_i = \text{probabilities: } f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$$

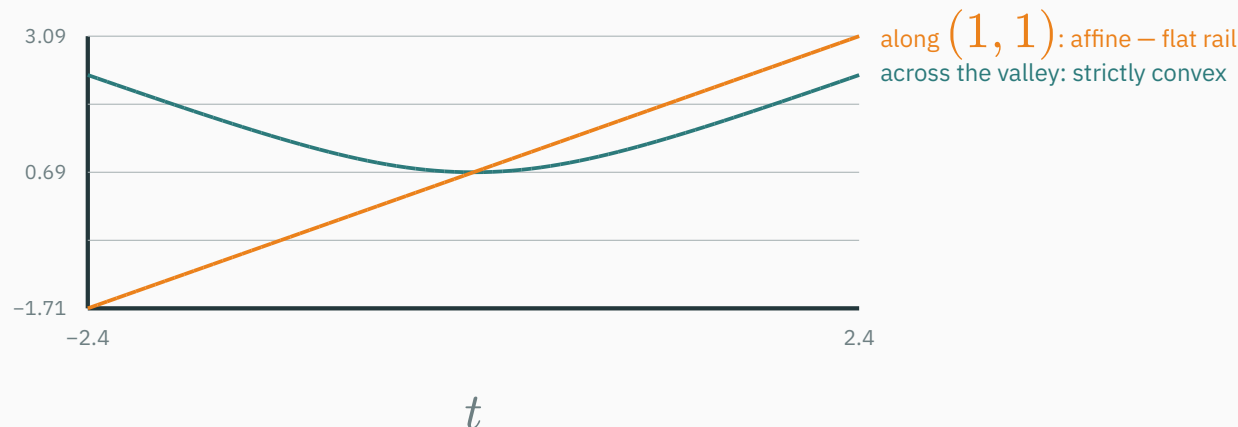
- For **concave** f , flip: $\mathbb{E}[\log X] \leq \log \mathbb{E}[X]$ — the single most-used inequality in probabilistic ML
- Module 5 runs on it: entropy bounds, $\text{KL} \geq 0$, the ELBO — all one chord, industrialized (L22–L24)

Weights on the simplex — the convex set from today's gallery. The definitions keep meeting.

☆☆☆ LSE's flat valley is L2's shift-invariance

★ OPTIONAL

Log-sum-exp is convex but not **strictly**: its bowl has one perfectly flat rail.



- Along $\mathbf{1} = (1, \dots, 1)$: $\text{LSE}(\mathbf{x} + c\mathbf{1}) = \text{LSE}(\mathbf{x}) + c$ — affine, zero curvature; indeed $H\mathbf{1} = (\text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top)\mathbf{1} = \mathbf{p} - \mathbf{p} = \mathbf{0}$
- That flat direction **is** L2's softmax shift-invariance — subtract-the-max just slides along the rail where nothing changes. Geometry and numerics, one fact.

Convexity is a promise: what you see locally is the truth globally.

Next: **Gradient Descent** — now we actually walk downhill: trust the linear model for one small step, and repeat.